

Nora Shao

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Education

University of British Columbia, BAsC in Engineering Physics Graduating April 2027

- Trek Excellence Scholarship 2022, 2023 (merit-based award offered to the top 5% of undergraduate students), Thomas Beeching Scholarship, Ausenco Scholarship for Women in Engineering

Skills

C, C++, Rust, Python, Java, CAN/SPI/I2C/UART/IEEE 802.3bt protocol, protobuf, FreeRTOS, MATLAB, Linux, PlatformIO, STM32CubeIDE/CubeMX, Intel Quartus Prime, Git, Altium Designer, LTSpice

Experience

Embedded Software Engineering Intern, Fab2 - San Francisco, CA May 2026 – Sept 2026

- Developed an Equivalent-Time Sampling (ETS) firmware pipeline on an STM32G474 to digitize and process 13.56MHz RF generator voltage and current signals, using the HRTIM peripheral and interrupts to micro-step sampling delays at rates beyond the ADC's Nyquist limit
- Wrote FreeRTOS-managed DSP tasks to filter signal outliers and calculate signal metrics (i.e. voltage/current at different harmonics, VI phase, forward/reflected power) on-chip using a Goertzel DFT algorithm
- Implemented multi-bus telemetry (CAN FD, USB Serial, SPI) on protobuf to transmit calculated metrics, and built a companion desktop visualization app in Rust to display real-time data during bring-up and validation
- Collaborated in a Git-based workflow: developing features on branches, opening/reviewing pull requests, and iterating through code reviews to merge firmware changes into the main branch

Electrical Engineering Intern, Motorola Solutions - Vancouver, BC May 2025 – Dec 2025

- Reverse-engineered and documented the C/C++ iris control firmware on an ATSAM20 microcontroller, deriving the software PI controller's closed-loop transfer function to characterize system response and stability
- Tuned a 14-parameter PTZ motor controller using a nested control architecture (open-loop feed-forward position control with inner closed-loop velocity control) to improve motion stability

Electrical & Firmware Subteam Member, Formula UBC Racing Design Team Sept 2022 – present

- Led a team of 6 members in designing and integrating critical electrical modules with a race car, and enabling suspension/powertrain data collection through AIM EVO5 Datalogger and in-house wiring harnesses
- Redesigned firmware for an F1-style steering wheel, programming STM32F405 MCU to add safety functionality (i.e. a gear shifting lock-out system that prevents downshifting at unsafe speeds by monitoring RPM data over CAN bus from ECU Haltech, and a filter for event-triggered module failure messages from in-house Power Distribution Module (PDM))

Technical Projects

FSAE Dash, Formula UBC Racing Design Team, *Github* September 2025 - May 2026

- Designed a 4-layer embedded PCB in Altium Designer around an STM32F405 microcontroller with CAN, SPI, I2C, USB, and power management circuitry
- Developed embedded firmware in C for the STM32F405 on STM32CubeIDE to receive Haltech ECU telemetry over CAN, display real-time vehicle data on MAX7219-driven 7-segment displays, and control auxiliary vehicle systems including the starter, ECU, cooling fans, and water pump
- Wrote hardware drivers for MAX7219 display controllers, WS2811 RGB LEDs, and an I2C LED driver to provide configurable shift lights and driver feedback

RTOS Scheduling Algorithms, *Github* Jan 2026 - April 2026

- Implemented SRP and CBS-based binary semaphore support in FreeRTOS for resource-sharing real-time tasks, including blocking-time-aware scheduling behavior and stack-sharing support to reduce memory overhead
- Modified the kernel and demo infrastructure to expose real-time behavior via hardware on an RP2040, using an AD2 logic analyzer to test the scheduling and resource-access behavior